Predicting Powerlifting Total from Bodyweight, Age, Sex, and Equipment Secondary Analysis of DOTS

By Abdul Ahmad and Edwin David

Predicting Powerlifting Total from Bodyweight, Age, Sex, and Equipment
Secondary Analysis of DOTS

CMPUT 195 Mini Project - By Edwin and Abdul

Main Goal: Predict competition TotalKg lifted using: Bodyweight, Age, Sex, and Equipment

- ↳ Secondary goal: Use DOTS to compare absolute vs relative strength

Dataset and Research Question

DATASET

Dataset: OpenPowerlifting public bulk CSV

Raw dataset: **3,891,860** rows over **42** columns

Key variables used: TotalKg, BodyweightKg, Age, Sex, Equipment, and DOTS

Research Question

Main question: How accurately can competition total be predicted using bodyweight, age, sex, and equipment?

Secondary question: How do these variables relate to relative strength as measured by DOTS?

Hypothesis and Project Logic

Hypothesis

- Bodyweight will have the strongest positive relationship with TotalKg
- Sex and equipment will also meaningfully affect TotalKg
- Age may matter, but less strongly
- DOTS will show that absolute and relative strength are related, but not identical

Why this project makes sense

- TotalKg is a direct performance outcome
- Regression is appropriate for a numerical target
- DOTS helps interpret pound-for-pound performance

Data Cleaning and Preprocessing

Pipeline	
Raw dataset	**3,891,860 rows**
Cleaned working dataset	**2,147,935 rows**
Model-ready dataset	**1,470,898 rows**

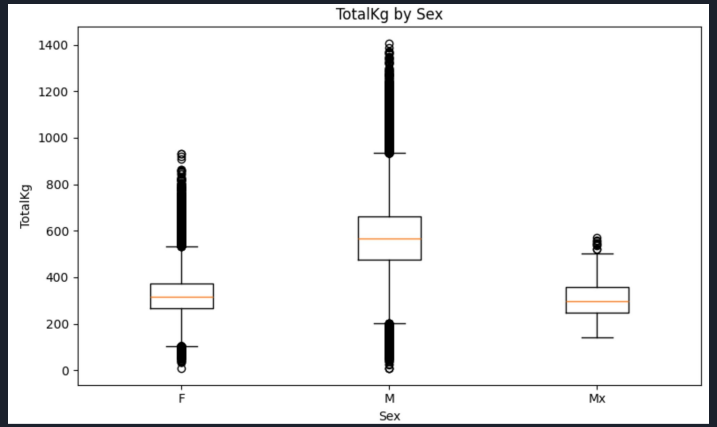
What we did

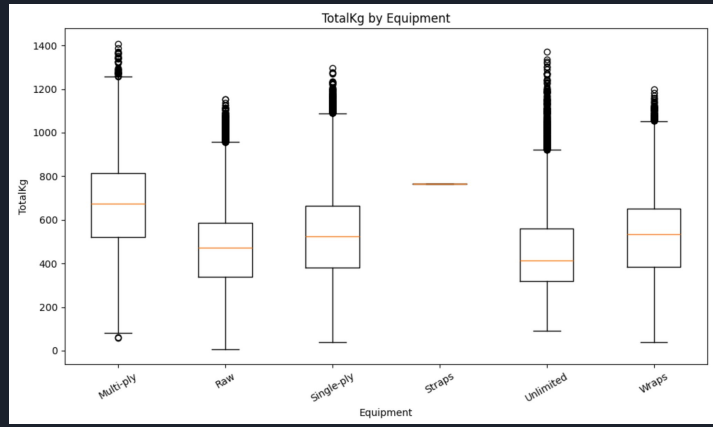
- Filtered to comparable full-powerlifting records
- Removed duplicate rows
- Filtered invalid/unusable rows
- Kept comparable full-powerlifting records
- Dropped rows with missing Age for final modeling
- One-hot encoded Sex and Equipment
- Standardized BodyweightKg and Age

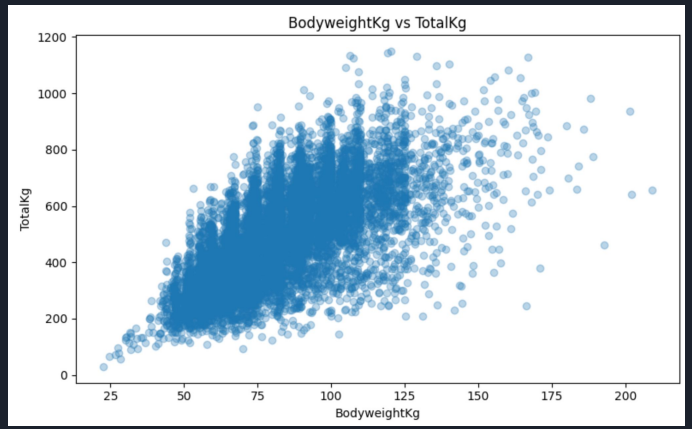
Exploratory Data Analysis

Main EDA findings

- TotalKg differs clearly across sex categories
- TotalKg also differs across equipment categories
- BodyweightKg has a strong positive relationship with TotalKg
- Age has a much weaker linear relationship with TotalKg







Key correlation values

- TotalKg & BodyweightKg: **0.657**
- TotalKg & Age: **0.041**
- TotalKg & Dots: **0.781**

Model Choice

Main model: Multiple Linear Regression

Target variable:

- TotalKg

Predictors:

- BodyweightKg
- Age
- Sex
- Equipment

Why this model?

- Numerical target
- Interpretable coefficients
- Easy to compare training vs test performance
- Easy to evaluate on training and test sets

Model Performance

Training vs Test Results		
Metric	**Train**	**Test**
RMSE	104.186	104.421
R^2	0.633	0.633

Interpretation

- Explains about 63.3% of the variation in TotalKg
- Error is about 104 kg
- Train and test performance are very similar
- Model generalizes consistently

Interpretation, Limitations, and Future Work

Main conclusion

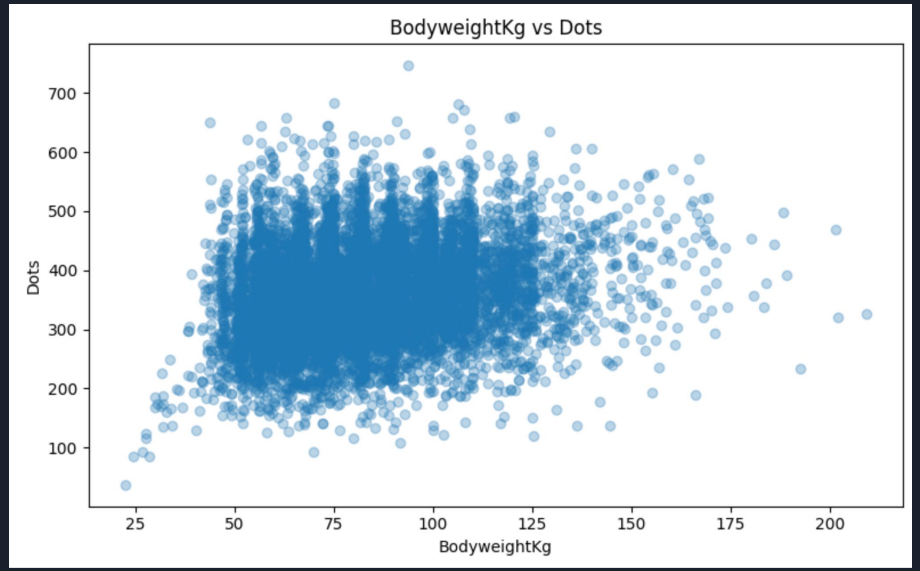
- The model was reasonably successful
- BodyweightKg was the strongest predictor
- Sex and Equipment also mattered
- DOTS helped compare relative vs absolute strength

Future work

- Compare models with and without Age
- Add competition history, federation, or lift-level variables
- Try subgroup-specific models

Strengths

- Interpretable model
- Very large dataset
- Stable train/test performance

Limitations

- Many rows missing Age
- Observational data, not causal
- Important factors were not included